

NATURE FOUNDATION MODELS

Building Foundation Models for Natural Systems

Phase 1: Decision Biology
Information-theoretic AI for cellular decision-making under uncertainty

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Nature Foundation Models

Scientific Intelligence for Natural Systems

Science Still Lacks Adaptive Intelligence

Despite advances in AI and computation:

- Scientific knowledge remains fragmented across scales and disciplines
- Discovery workflows are still slow, empirical, and trial-and-error
- Existing AI systems predict patterns but struggle with causal reasoning
- We still lack adaptive systems that can model and reason about nature

The Opportunity

Build the scientific intelligence layer for natural systems.

A new class of models that can:

- learn causal dynamics
- simulate complex adaptive systems
- reason under uncertainty
- generate and test scientific hypotheses
- unify reasoning across biology, chemistry, physics, and environmental systems

From predictive AI → adaptive scientific intelligence

The Insight

Biology Is an Adaptive Information System

Cells do not simply react.

They:

- process signals
- adapt under uncertainty
- respond to perturbations
- coordinate through signaling networks

The Problem

Current AI systems:

- predict patterns
- interpolate across datasets
- struggle with causal reasoning
- cannot explain hidden mechanisms

Scientific discovery requires:

- experimentation
- intervention analysis
- adaptive reasoning

Science needs systems that can reason – not just predict

ASRA + Decision Biology

Our Architecture

ASRA

Adaptive Scientific Reasoning Architecture

A reasoning engine for:

- world models
- causal inference
- adaptive experimentation
- hidden-mechanics discovery

Decision Biology

The first scientific domain built on ASRA.

Treating:

- perturbations as interventions
- signaling pathways as world models
- cellular adaptation as causal reasoning

From predictive AI → adaptive scientific intelligence

Why Science Needs World Models

Existing AI Systems

Most current systems are optimized for:

- pattern prediction
- static datasets
- fixed objectives
- known action spaces
- task-specific learning

They perform well in familiar environments –
but struggle with adaptive scientific reasoning.

Scientific Environments Are Different

Scientific systems require:

- experimentation
- causal reasoning
- uncertainty reduction
- intervention analysis
- hidden-mechanics discovery

Scientific intelligence requires more than prediction

Why ASRA Is Different

ASRA Is Built for Adaptive Reasoning

Designed for:

- intervention-driven learning
- world-model construction
- causal inference
- hidden-mechanics discovery
- adaptive experimentation

Long-Term Goal

AI systems that behave less like static predictors – and more like scientific investigators.

Platform Hierarchy

ASRA

→ adaptive reasoning engine

Decision Biology

→ first scientific domain

Nature Foundation Models

→ multi-domain scientific intelligence platform

Toward adaptive scientific intelligence